ALERT SYSTEM IMPLEMENTATION

Testing Scenarios and Results

HealthTrack: Remote Monitoring for Home-Based Cardiac Rehabilitation

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Summary

Database	Tests	Passed	Failed	Errors	Skipped	Duration (s)
sqlite	305	305	0	0	0	10.70

Line coverage of `app/`: **96.1%** (3377 of 3515 statements). Target from requirement NFR-16: 80%.

Testing scenarios

52 named scenarios (55 test cases), 52 passed. Each scenario is an automated test (a parametrized scenario runs once per case); the requirement column refers to the Phase 1 requirements matrix.

ID	Scenario	Requirement	Result	Test
AS-01	Exercise heart rate above the prescribed ceiling on 2 consecutive readings raises a critical alert and queues dashboard, SMS, and email for the primary clinician	FR-10, FR-12, FR-13	PASS	`test_critical_hr_above_prescribed_ceiling`
AS-02	The same heart rate does not alert for a patient whose prescribed ceiling is higher	FR-10	PASS	`test_same_value_patient_specific`
AS-03	A single breaching reading does not alert when the rule needs consecutive readings	FR-10	PASS	`test_single_spike_filtered`
AS-04	Breaching readings farther apart than the rule window are not consecutive	FR-10	PASS	`test_consecutive_window`
AS-05	Context filters: a resting reading never triggers the exercise ceiling rule, but two high resting readings trigger the resting rule	FR-10	PASS	`test_context_filters`
AS-06	Suspect readings (implausible values) are stored but never alert	FR-07, FR-10	PASS	`test_suspect_never_alerts`
AS-07	A patient override (beta-blocker, resting limit 90) takes precedence over the program default	FR-10	PASS	`test_override_precedence`
AS-08	Without a prescription the program default applies and is labeled as the source	FR-10	PASS	`test_default_without_prescription`
AS-09	SpO2 below the prescription floor raises a critical alert in any context	FR-10	PASS	`test_spo2_low`
AS-10	Blood pressure rules: exercise systolic above the prescribed maximum is high; resting systolic at or above 160 is moderate	FR-10	PASS	`test_bp_rules`
AS-11	Missed sessions: no exercise session for a week (3 prescribed per week) raises a high alert; the adherent control stays silent	FR-11	PASS	`test_missed_sessions`
AS-12	Minutes in the prescribed heart rate zone falling at least 25% week over week raises a moderate alert; the control stays silent	FR-11	PASS	`test_zone_minutes_decline`

ID	Scenario	Requirement	Result	Test
AS-13	7-day resting heart rate more than 8 bpm above the prescription baseline raises a moderate alert; the control stays silent	FR-11	PASS	`test_resting_hr_rise`
AS-14	Average daily steps falling at least 30% from the week before raises a moderate alert; the control stays silent	FR-11	PASS	`test_steps_decline`
AS-15	No readings for 48 hours raises a low-severity data-gap alert; recent data stays silent	FR-11	PASS	`test_data_gap`
AS-16	Too little history never looks like a decline: trends need 4+ days in each week and a minimum baseline	FR-11	PASS	`test_insufficient_data`
AS-17	Repeated breaches attach to the open alert instead of creating new alerts or notifications	FR-12	PASS	`test_repeats_absorbed`
AS-18	After resolution, the same moderate rule inside its suppression window is logged as suppressed with no notifications; after the window it alerts again	FR-12	PASS	`test_suppression_window`
AS-19	Critical alerts are never suppressed: a repeat right after resolution notifies again	FR-12	PASS	`test_critical_never_suppressed`
AS-20	An active high-severity alert masks a lower-severity alert on the same measurement	FR-12	PASS	`test_masking`
AS-21	Low-severity alerts reach the dashboard now and email in the next morning digest	FR-12	PASS	`test_low_goes_to_digest`
AS-22	Alert-storm cap: past the hourly limit, a patient's non-critical email and SMS go to the digest while the dashboard is still notified immediately	FR-12	PASS	`test_storm_cap`
AS-23	Critical notifications are dispatched before lower-priority ones that were queued earlier	FR-12	PASS	`test_critical_first`
AS-24	Notification payload carries alert id, patient id, severity, rule, triggering values, and time; email and SMS text carry no patient name, MRN, or measured values	FR-13, NFR-06	PASS	`test_payload_and_minimum_necessary`
AS-25	Dispatch sends every due message, logs a 'notified' history event per delivery, and writes email and SMS to the outbox when no gateway is configured	FR-13	PASS	`test_dispatch_logs_delivery`
AS-26	A failing channel retries with backoff (15 s, then 30 s) and is marked failed with a reason after 3 attempts, without blocking other channels	FR-13	PASS	`test_retry_then_fail`
AS-27	Low-severity email is held for the morning digest and several alerts go out as one message per recipient	FR-12	PASS	`test_digest_groups_messages`
AS-28	SMS needs a phone number on the recipient account, and the digest message lists each alert with its sign-in link	FR-13	PASS	`test_sms_requires_phone_and_digest_text`
AS-29	A clinician on the care team acknowledges an alert; who and when are recorded in the alert history and the audit log	FR-14, NFR-05	PASS	`test_acknowledge`
AS-30	Patients and administrators cannot acknowledge (403); a clinician outside the care team gets 404 and the attempt is audited	FR-14, NFR-04	PASS	`test_ack_permissions[admin-403]`
AS-30	Patients and administrators cannot acknowledge (403); a clinician outside the care team gets 404 and the attempt is audited	FR-14, NFR-04	PASS	`test_ack_permissions[it-403]`

ID	Scenario	Requirement	Result	Test
AS-30	Patients and administrators cannot acknowledge (403); a clinician outside the care team gets 404 and the attempt is audited	FR-14, NFR-04	PASS	`test_ack_permissions [marcus-404]`
AS-30	Patients and administrators cannot acknowledge (403); a clinician outside the care team gets 404 and the attempt is audited	FR-14, NFR-04	PASS	`test_ack_permissions [robert_login-403]`
AS-31	Resolution needs a note, records acknowledgment if missing, and closed alerts reject further actions	FR-14	PASS	`test_resolution_rules`
AS-32	An unacknowledged critical alert escalates to the supervising physician at 15 minutes (not before), to the whole care team on every channel at 30 minutes, and stops at the maximum level	FR-15	PASS	`test_escalation_ladder`
AS-33	Acknowledging an alert stops escalation	FR-14, FR-15	PASS	`test_ack_stops_escalation`
AS-34	Moderate alerts escalate once (after 4 hours); low alerts never escalate	FR-15	PASS	`test_policy_levels`
AS-35	History search filters by patient, severity, status, rule, date range, and acknowledging user, with correct paging	FR-16	PASS	`test_history_search`
AS-36	Alert detail returns the full timeline (created, notified, acknowledged, resolved) and minutes from trigger to acknowledgment and resolution	FR-16	PASS	`test_detail_timeline`
AS-37	Metrics report counts by severity and status and minutes to acknowledge, for clinicians (their team) and admins (whole program)	FR-16, FR-26	PASS	`test_metrics`
AS-38	Only a physician changes program rules, the reason is audited, and a critical rule cannot be given a suppression window	FR-12, NFR-05	PASS	`test_rule_changes`
AS-39	A physician issues a new prescription version; the next reading is judged against the new ceiling	FR-02, FR-10	PASS	`test_prescription_endpoint`
AS-40	Patient-specific thresholds need a clinical reason, are refused for prescription-driven rules, and removing one falls back to the default	FR-10	PASS	`test_patient_thresholds`
AS-41	Escalation policies are validated: critical must escalate within 30 minutes, only low may use the digest, and the dashboard channel is required	FR-15	PASS	`test_policy_updates`
AS-42	A dashboard connected with a ticket receives the new readings and the critical alert pushed over the websocket within the 5-second target	FR-18, NFR-07	PASS	`test_push_reading_and_alert`
AS-43	Tickets are single-use; a reused or invalid ticket and a wrong origin are refused	NFR-03	PASS	`test_ticket_rules`
AS-44	Events are filtered by care team: a clinician never receives another team's patient	NFR-04	PASS	`test_scope_filtering`
AS-45	Subscribing to one patient narrows reading events, while alerts still arrive for every patient in scope	FR-18	PASS	`test_subscription_narrows_vitals`
AS-46	Dashboard notifications go only to their recipient's open sessions	FR-13	PASS	`test_notification_goes_to_recipient`
AS-47	A device batch that crosses the ceiling returns alerts_raised and the alert appears on the care team's worklist, with repeats folded into it	FR-10, FR-18	PASS	`test_batch_raises_alert`

ID	Scenario	Requirement	Result	Test
AS-48	If alert evaluation fails, the reading is still stored and the request succeeds	NFR-12	PASS	`test_evaluation_failure_never_loses_readings`
AS-49	The monitoring worklist ranks the patient with a critical alert first and carries latest values and prescription limits	FR-17, NFR-15	PASS	`test_worklist`
AS-50	Uploading a day of activity re-checks trend rules, so a step decline alerts without waiting for the scheduled job	FR-11	PASS	`test_activity_upload_triggers_patterns`
AS-51	Migration 0003 adds the alerting tables with the 12 default rules and 4 escalation policies to an existing phase 3 database, and downgrades cleanly	NFR-17	PASS	`test_schema_update_0003_adds_alerting`
AS-52	Background workers send queued notifications as soon as they are woken, without waiting for the poll interval	NFR-07	PASS	`test_workers_dispatch_on_wake`

Results by area (sqlite)

Test module	What it covers	Passed	Failed	Skipped
`test_models.py`	SQLAlchemy models: field constraints, foreign keys, invalid data	68	0	0
`test_repositories.py`	Data access layer: CRUD, filtering, pagination, errors, logging	50	0	0
`test_schemas_security.py`	Request validation schemas and security primitives	40	0	0
`test_migrations.py`	Alembic migrations: create, schema update, downgrade, drift	4	0	0
`test_api_auth.py`	Auth endpoints: login, lockout, MFA, refresh rotation, logout	13	0	0
`test_api_patients.py`	Patient endpoints: CRUD, search, role and scope matrix, audit	29	0	0
`test_api_monitoring.py`	Vital sign, activity, and device endpoints; error handling	34	0	0
`test_alert_engine.py`	Alert engine: thresholds, patient limits, fatigue controls, escalation	24	0	0
`test_alert_patterns.py`	Activity pattern detection against a declining patient and a stable control	9	0	0
`test_notifications.py`	Notification payloads, channels, delivery logging, retries, digests	7	0	0
`test_api_alerts.py`	Alert API: ingestion integration, worklist, acknowledgment, roles, history, configuration	19	0	0
`test_realtime.py`	Live stream: websocket auth, push latency, scope filtering, workers	8	0	0

Failures

None. Every test passed.

Coverage by module

Module	Statements	Missed	Coverage
`app/__init__.py`	1	0	100%
`app/alerts/__init__.py`	0	0	100%
`app/alerts/defaults.py`	18	0	100%

Module	Statements	Missed	Coverage
app/alerts/engine.py	211	7	97%
app/alerts/notifications.py	153	19	88%
app/alerts/patterns.py	109	5	95%
app/alerts/service.py	51	4	92%
app/alerts/workers.py	83	24	71%
app/api/__init__.py	0	0	100%
app/api/deps.py	60	2	97%
app/api/handlers.py	41	1	98%
app/api/responses.py	15	0	100%
app/api/routes/__init__.py	0	0	100%
app/api/routes/activity.py	60	1	98%
app/api/routes/alert_config.py	74	1	99%
app/api/routes/alerts.py	110	2	98%
app/api/routes/auth.py	85	1	99%
app/api/routes/patients.py	52	0	100%
app/api/routes/stream.py	134	10	93%
app/api/routes/vitals.py	88	1	99%
app/config.py	55	1	98%
app/db.py	47	14	70%
app/errors.py	31	0	100%
app/logging_config.py	36	2	94%
app/main.py	76	3	96%
app/models/__init__.py	7	0	100%
app/models/alerting.py	136	0	100%
app/models/audit.py	27	1	96%
app/models/base.py	31	0	100%
app/models/enums.py	108	0	100%
app/models/monitoring.py	72	0	100%
app/models/patient.py	86	0	100%
app/models/user.py	43	2	95%
app/realtime/__init__.py	0	0	100%
app/realtime/broker.py	76	5	93%
app/realtime/tickets.py	29	0	100%
app/repositories/__init__.py	5	0	100%
app/repositories/alerts.py	221	17	92%
app/repositories/audit.py	34	0	100%

Module	Statements	Missed	Coverage
app/repositories/base.py	75	6	92%
app/repositories/monitoring.py	226	2	99%
app/repositories/patients.py	98	2	98%
app/repositories/users.py	62	3	95%
app/schemas/__init__.py	0	0	100%
app/schemas/alerts.py	214	0	100%
app/schemas/auth.py	24	0	100%
app/schemas/common.py	30	0	100%
app/schemas/monitoring.py	128	0	100%
app/schemas/patient.py	110	1	99%
app/security.py	83	1	99%

Slowest tests

Test	Seconds
test_api_monitoring::test_openapi_documents_errors_and_security	0.485
test_migrations::test_downgrade_to_base_and_back	0.302
test_migrations::test_upgrade_head_matches_models	0.189
test_migrations::test_schema_update_0003_adds_alerting	0.119
test_api_alerts::test_history_search	0.117
test_api_alerts::test_batch_raises_alert	0.108
test_realtime::test_subscription_narrows_vitals	0.104
test_api_alerts::test_jobs_endpoint_and_notifications	0.102
test_api_monitoring::test_filters	0.102
test_alert_patterns::test_repeated_pattern_runs_do_not_duplicate	0.101

Complete list

test_models.py

- PASS TestPatientConstraints::test_valid_patient_persists_with_defaults
- PASS TestPatientConstraints::test_duplicate_mrn_rejected
- PASS TestPatientConstraints::test_malformed_mrn_rejected_by_model_validator[ht-1]
- PASS TestPatientConstraints::test_malformed_mrn_rejected_by_model_validator[HT 100001]
- PASS TestPatientConstraints::test_malformed_mrn_rejected_by_model_validator[XXXXXXXXXXXXXXXXXXXX]
- PASS TestPatientConstraints::test_malformed_mrn_rejected_by_model_validator[]
- PASS TestPatientConstraints::test_format_validators[phone-7195550111]
- PASS TestPatientConstraints::test_format_validators[postal_code-8092]

- PASS TestPatientConstraints::test_format_validators[email-nope]
- PASS TestPatientConstraints::test_email_is_normalized_to_lowercase
- PASS TestPatientConstraints::test_minor_at_qualifying_event_rejected
- PASS TestPatientConstraints::test_future_qualifying_event_rejected
- PASS TestPatientConstraints::test_event_after_referral_rejected
- PASS TestPatientConstraints::test_enrollment_before_referral_rejected
- PASS TestPatientConstraints::test_active_without_enrollment_date_rejected
- PASS TestPatientConstraints::test_referred_without_enrollment_date_allowed
- PASS TestPatientConstraints::test_graduated_requires_graduation_date
- PASS TestPatientConstraints::test_graduation_date_without_graduated_status_rejected
- PASS TestPatientConstraints::test_active_high_risk_rejected
- PASS TestPatientConstraints::test_paused_high_risk_allowed
- PASS TestPatientConstraints::test_target_sessions_range[0]
- PASS TestPatientConstraints::test_target_sessions_range[73]
- PASS TestPatientConstraints::test_blank_name_rejected
- PASS TestPatientConstraints::test_invalid_enum_value_rejected
- PASS TestCareTeam::test_second_current_primary_clinician_rejected
- PASS TestCareTeam::test_ended_assignment_does_not_block_new_primary
- PASS TestCareTeam::test_covering_clinicians_may_overlap
- PASS TestCareTeam::test_end_before_start_rejected
- PASS TestVitalSignConstraints::test_valid_reading_persists
- PASS TestVitalSignConstraints::test_hard_limits[kw0]
- PASS TestVitalSignConstraints::test_hard_limits[kw1]
- PASS TestVitalSignConstraints::test_hard_limits[kw2]
- PASS TestVitalSignConstraints::test_hard_limits[kw3]
- PASS TestVitalSignConstraints::test_hard_limits[kw4]
- PASS TestVitalSignConstraints::test_hard_limits[kw5]
- PASS TestVitalSignConstraints::test_reading_without_measurement_rejected
- PASS TestVitalSignConstraints::test_bp_must_be_paired
- PASS TestVitalSignConstraints::test_systolic_must_exceed_diastolic
- PASS TestVitalSignConstraints::test_manual_reading_requires_author
- PASS TestVitalSignConstraints::test_duplicate_idempotency_key_rejected
- PASS TestVitalSignConstraints::test_same_key_allowed_for_different_patients
- PASS TestVitalSignConstraints::test_naive_datetime_rejected
- PASS TestVitalSignConstraints::test_datetimes_round_trip_as_utc
- PASS TestVitalSignConstraints::test_quality_assessment[args0-valid]
- PASS TestVitalSignConstraints::test_quality_assessment[args1-suspect]
- PASS TestVitalSignConstraints::test_quality_assessment[args2-suspect]
- PASS TestVitalSignConstraints::test_quality_assessment[args3-suspect]
- PASS TestVitalSignConstraints::test_quality_assessment[args4-suspect]
- PASS TestVitalSignConstraints::test_quality_assessment[args5-suspect]

- PASS TestVitalSignConstraints::test_quality_assessment[args6-valid]
- PASS TestActivityAndDevices::test_valid_day_persists
- PASS TestActivityAndDevices::test_one_row_per_patient_per_day
- PASS TestActivityAndDevices::test_activity_limits[kw0]
- PASS TestActivityAndDevices::test_activity_limits[kw1]
- PASS TestActivityAndDevices::test_activity_limits[kw2]
- PASS TestActivityAndDevices::test_activity_limits[kw3]
- PASS TestActivityAndDevices::test_activity_limits[kw4]
- PASS TestActivityAndDevices::test_activity_limits[kw5]
- PASS TestActivityAndDevices::test_active_device_requires_patient
- PASS TestActivityAndDevices::test_duplicate_serial_rejected
- PASS TestForeignKeys::test_vital_for_missing_patient_rejected
- PASS TestForeignKeys::test_activity_for_missing_patient_rejected
- PASS TestForeignKeys::test_care_team_for_missing_user_rejected
- PASS TestForeignKeys::test_vital_with_unknown_device_rejected
- PASS TestForeignKeys::test_relationships_navigate_both_ways
- PASS TestForeignKeys::test_deleting_patient_cascades_to_monitoring_data
- PASS TestForeignKeys::test_user_who_entered_readings_cannot_be_deleted
- PASS TestAuditLog::test_audit_rows_cannot_be_updated_or_deleted

test_repositories.py

- PASS TestPatientRepository::test_create_adds_two_care_team_members
- PASS TestPatientRepository::test_create_rejects_non_physician_supervisor
- PASS TestPatientRepository::test_create_rejects_staff_account_as_patient_login
- PASS TestPatientRepository::test_create_rejects_inactive_clinician
- PASS TestPatientRepository::test_duplicate_mrn_becomes_conflict_error
- PASS TestPatientRepository::test_check_violation_becomes_validation_error_and_session_survives
- PASS TestPatientRepository::test_model_validator_error_becomes_validation_error
- PASS TestPatientRepository::test_get_missing_raises_not_found
- PASS TestPatientRepository::test_list_scoped_to_clinician_care_team
- PASS TestPatientRepository::test_future_dated_assignment_grants_no_access
- PASS TestPatientRepository::test_list_filters_status_sort_and_paginates
- PASS TestPatientRepository::test_search_matches_name_or_mrn_case_insensitively
- PASS TestPatientRepository::test_update_reports_changed_columns_only
- PASS TestPatientRepository::test_soft_delete_hides_and_withdraws
- PASS TestPatientRepository::test_hard_delete_logs_warning
- PASS TestPatientRepository::test_create_is_logged
- PASS TestVitalSignRepository::test_create_assesses_quality
- PASS TestVitalSignRepository::test_manual_reading_records_author
- PASS TestVitalSignRepository::test_manual_reading_without_author_rejected
- PASS TestVitalSignRepository::test_device_reading_requires_device

- PASS TestVitalSignRepository::test_device_must_belong_to_patient
- PASS TestVitalSignRepository::test_inactive_device_rejected
- PASS TestVitalSignRepository::test_create_updates_device_last_sync
- PASS TestVitalSignRepository::test_duplicate_idempotency_key_is_conflict
- PASS TestVitalSignRepository::test_bulk_create_counts_duplicates_suspect_and_rejected
- PASS TestVitalSignRepository::test_bulk_create_accepts_mixed_measurement_types
- PASS TestVitalSignRepository::test_bulk_create_rejects_manual_rows
- PASS TestVitalSignRepository::test_list_filters_by_date_range_context_metric_quality
- PASS TestVitalSignRepository::test_end_is_exclusive
- PASS TestVitalSignRepository::test_invalid_ranges_rejected[delta0-after start]
- PASS TestVitalSignRepository::test_invalid_ranges_rejected[delta1-92 days]
- PASS TestVitalSignRepository::test_latest_ignores_suspect_readings
- PASS TestVitalSignRepository::test_daily_summary_aggregates_valid_readings_per_day
- PASS TestVitalSignRepository::test_update_only_context_and_quality
- PASS TestVitalSignRepository::test_delete_and_get_missing
- PASS TestActivityRepository::test_upsert_creates_then_replaces
- PASS TestActivityRepository::test_upsert_rejects_future_date_and_foreign_device
- PASS TestActivityRepository::test_check_violation_translated
- PASS TestActivityRepository::test_list_range_summary_and_delete
- PASS TestActivityRepository::test_empty_summary
- PASS TestDeviceRepository::test_register_list_get
- PASS TestDeviceRepository::test_duplicate_serial_conflict
- PASS TestUserAndAuditRepositories::test_email_lookup_is_case_insensitive
- PASS TestUserAndAuditRepositories::test_failed_logins_lock_account
- PASS TestUserAndAuditRepositories::test_sessions_revoke
- PASS TestUserAndAuditRepositories::test_duplicate_email_conflict
- PASS TestUserAndAuditRepositories::test_get_missing_user
- PASS TestUserAndAuditRepositories::test_audit_record_and_filter
- PASS TestErrorTranslation::test_operational_error_becomes_503_and_is_logged
- PASS TestErrorTranslation::test_unknown_foreign_key_message

test_schemas_security.py

- PASS TestPatientSchemas::test_valid
- PASS TestPatientSchemas::test_invalid[override0]
- PASS TestPatientSchemas::test_invalid[override1]
- PASS TestPatientSchemas::test_invalid[override2]
- PASS TestPatientSchemas::test_invalid[override3]
- PASS TestPatientSchemas::test_invalid[override4]
- PASS TestPatientSchemas::test_invalid[override5]
- PASS TestPatientSchemas::test_invalid[override6]
- PASS TestPatientSchemas::test_invalid[override7]

- PASS TestPatientSchemas::test_invalid[override8]
- PASS TestPatientSchemas::test_invalid[override9]
- PASS TestPatientSchemas::test_invalid[override10]
- PASS TestPatientSchemas::test_invalid[override11]
- PASS TestPatientSchemas::test_update_is_partial_and_checks_date_order
- PASS TestMonitoringSchemas::test_valid
- PASS TestMonitoringSchemas::test_invalid[override0]
- PASS TestMonitoringSchemas::test_invalid[override1]
- PASS TestMonitoringSchemas::test_invalid[override2]
- PASS TestMonitoringSchemas::test_invalid[override3]
- PASS TestMonitoringSchemas::test_invalid[override4]
- PASS TestMonitoringSchemas::test_invalid[override5]
- PASS TestMonitoringSchemas::test_invalid[override6]
- PASS TestMonitoringSchemas::test_invalid[override7]
- PASS TestMonitoringSchemas::test_invalid[override8]
- PASS TestMonitoringSchemas::test_small_clock_skew_allowed
- PASS TestMonitoringSchemas::test_batch_size_limits
- PASS TestMonitoringSchemas::test_activity_minutes_fit_in_day
- PASS TestPasswords::test_hash_and_verify
- PASS TestPasswords::test_hashes_are_salted
- PASS TestPasswords::test_overlong_password_refused
- PASS TestTokens::test_round_trip
- PASS TestTokens::test_tampered_token_rejected
- PASS TestTokens::test_none_algorithm_rejected
- PASS TestTokens::test_expired_token_rejected
- PASS TestTokens::test_wrong_audience_rejected
- PASS TestTokens::test_refresh_tokens_are_random_and_hashed
- PASS TestTotpAndLimiter::test_rfc6238_reference_vector
- PASS TestTotpAndLimiter::test_verify_window
- PASS TestTotpAndLimiter::test_limiter_blocks_after_limit
- PASS test_unknown_email_dummy_hash_uses_configured_cost

test_migrations.py

- PASS test_upgrade_head_matches_models
- PASS test_schema_update_0002_adds_device_registry
- PASS test_downgrade_to_base_and_back
- PASS test_schema_update_0003_adds_alerting

test_api_auth.py

- PASS test_login_returns_tokens_and_profile
- PASS test_patient_refresh_lifetime_is_seven_days
- PASS test_bad_password_and_unknown_email_look_identical

- PASS test_five_failures_lock_the_account
- PASS test_inactive_account_cannot_log_in
- PASS test_mfa_required_when_enabled
- PASS test_login_is_audited_without_password
- PASS test_rate_limit_returns_429_with_retry_after
- PASS test_refresh_rotates_and_detects_reuse
- PASS test_refresh_with_unknown_token
- PASS test_logout_revokes_access_token_immediately
- PASS test_missing_garbage_and_stale_role_tokens
- PASS test_deactivated_user_token_stops_working

test_api_patients.py

- PASS TestCreate::test_clinician_creates_patient
- PASS TestCreate::test_duplicate_mrn_409
- PASS TestCreate::test_validation_errors_list_each_field
- PASS TestCreate::test_unknown_field_rejected
- PASS TestCreate::test_supervisor_must_be_physician
- PASS TestCreate::test_patients_and_it_cannot_create[robert_login]
- PASS TestCreate::test_patients_and_it_cannot_create[it]
- PASS TestReadAndScope::test_get_matrix[robert_login-robert-200]
- PASS TestReadAndScope::test_get_matrix[robert_login-linda-404]
- PASS TestReadAndScope::test_get_matrix[dana-robert-200]
- PASS TestReadAndScope::test_get_matrix[dana-thomas-404]
- PASS TestReadAndScope::test_get_matrix[marcus-thomas-200]
- PASS TestReadAndScope::test_get_matrix[marcus-robert-404]
- PASS TestReadAndScope::test_get_matrix[raman-thomas-200]
- PASS TestReadAndScope::test_get_matrix[admin-thomas-200]
- PASS TestReadAndScope::test_get_matrix[it-robert-403]
- PASS TestReadAndScope::test_out_of_scope_is_404_and_audited_as_denied
- PASS TestReadAndScope::test_missing_patient_404
- PASS TestReadAndScope::test_malformed_id_422
- PASS TestReadAndScope::test_list_is_scoped_and_paginated
- PASS TestReadAndScope::test_list_filters_and_sort
- PASS TestReadAndScope::test_page_size_capped
- PASS TestReadAndScope::test_patients_cannot_list
- PASS TestReadAndScope::test_search_in_body_and_scoped
- PASS TestUpdateDelete::test_clinician_updates_and_audit_stores_column_names_only
- PASS TestUpdateDelete::test_patient_may_edit_contact_fields_only
- PASS TestUpdateDelete::test_rule_violation_on_update_is_422
- PASS TestUpdateDelete::test_graduate_requires_date
- PASS TestUpdateDelete::test_only_admin_soft_deletes

test_api_monitoring.py

- PASS TestVitalCollection::test_patient_app_posts_reading
- PASS TestVitalCollection::test_suspect_value_stored_and_flagged
- PASS TestVitalCollection::test_invalid_readings_422_with_field[override0-heart_rate]
- PASS TestVitalCollection::test_invalid_readings_422_with_field[override1-context]
- PASS TestVitalCollection::test_invalid_readings_422_with_field[override2-recorded_at]
- PASS TestVitalCollection::test_bp_order_rule
- PASS TestVitalCollection::test_future_reading_rejected
- PASS TestVitalCollection::test_foreign_device_rejected
- PASS TestVitalCollection::test_duplicate_idempotency_key_409
- PASS TestVitalCollection::test_clinician_manual_entry_records_author
- PASS TestVitalCollection::test_patient_cannot_post_manual
- PASS TestVitalCollection::test_write_scope[robert_login-linda-404]
- PASS TestVitalCollection::test_write_scope[marcus-robert-404]
- PASS TestVitalCollection::test_write_scope[admin-robert-403]
- PASS TestVitalCollection::test_write_scope[it-robert-403]
- PASS TestVitalCollection::test_batch_upload_and_safe_replay
- PASS TestVitalCollection::test_batch_with_mixed_measurements
- PASS TestVitalCollection::test_batch_limits
- PASS TestVitalQueries::test_default_window_is_last_24_hours
- PASS TestVitalQueries::test_filters
- PASS TestVitalQueries::test_range_validation
- PASS TestVitalQueries::test_latest_and_daily_summary
- PASS TestVitalQueries::test_reads_are_audited_with_row_counts
- PASS TestVitalQueries::test_get_patch_delete_single_reading
- PASS TestActivityAndDevices::test_put_creates_201_then_replaces_200
- PASS TestActivityAndDevices::test_invalid_activity
- PASS TestActivityAndDevices::test_history_summary_and_scope
- PASS TestActivityAndDevices::test_delete_is_clinician_only
- PASS TestActivityAndDevices::test_devices
- PASS TestCrossCutting::test_health
- PASS TestCrossCutting::test_request_id_echoed_and_security_headers
- PASS TestCrossCutting::test_unknown_route_problem_json
- PASS TestCrossCutting::test_unhandled_error_is_500_without_internals
- PASS TestCrossCutting::test_openapi_documents_errors_and_security

test_alert_engine.py

- PASS test_critical_hr_above_prescribed_ceiling
- PASS test_same_value_patient_specific
- PASS test_single_spike_filtered
- PASS test_consecutive_window

- PASS test_context_filters
- PASS test_suspect_never_alerts
- PASS test_override_precedence
- PASS test_default_without_prescription
- PASS test_spo2_low
- PASS test_bp_rules
- PASS test_comparators_and_resolver_order
- PASS test_repeats_absorbed
- PASS test_suppression_window
- PASS test_critical_never_suppressed
- PASS test_masking
- PASS test_storm_cap
- PASS test_low_goes_to_digest
- PASS test_escalation_ladder
- PASS test_ack_stops_escalation
- PASS test_policy_levels
- PASS test_resolution_rules
- PASS test_alert_events_are_append_only
- PASS test_prescription_versions
- PASS test_override_rejected_for_prescription_rules

test_alert_patterns.py

- PASS test_missed_sessions
- PASS test_zone_minutes_decline
- PASS test_resting_hr_rise
- PASS test_steps_decline
- PASS test_data_gap
- PASS test_control_patient_completely_silent
- PASS test_insufficient_data
- PASS test_repeated_pattern_runs_do_not_duplicate
- PASS test_pattern_titles_are_readable

test_notifications.py

- PASS test_payload_and_minimum_necessary
- PASS test_dispatch_logs_delivery
- PASS test_retry_then_fail
- PASS test_digest_groups_messages
- PASS test_critical_first
- PASS test_dashboard_channel_targets_one_user
- PASS test_sms_requires_phone_and_digest_text

test_api_alerts.py

- PASS test_batch_raises_alert
- PASS test_evaluation_failure_never_loses_readings
- PASS test_worklist
- PASS test_acknowledge
- PASS test_ack_permissions[robert_login-403]
- PASS test_ack_permissions[admin-403]
- PASS test_ack_permissions[it-403]
- PASS test_ack_permissions[marcus-404]
- PASS test_resolve_via_api
- PASS test_history_search
- PASS test_detail_timeline
- PASS test_metrics
- PASS test_rule_changes
- PASS test_prescription_endpoint
- PASS test_patient_thresholds
- PASS test_policy_updates
- PASS test_activity_upload_triggers_patterns
- PASS test_jobs_endpoint_and_notifications
- PASS test_alert_status_enum_and_health

test_realtime.py

- PASS test_push_reading_and_alert
- PASS test_ticket_rules
- PASS test_access_token_auth_for_non_browser_clients
- PASS test_scope_filtering
- PASS test_subscription_narrows_vitals
- PASS test_notification_goes_to_recipient
- PASS test_broker_drops_oldest_for_slow_clients
- PASS test_workers_dispatch_on_wake